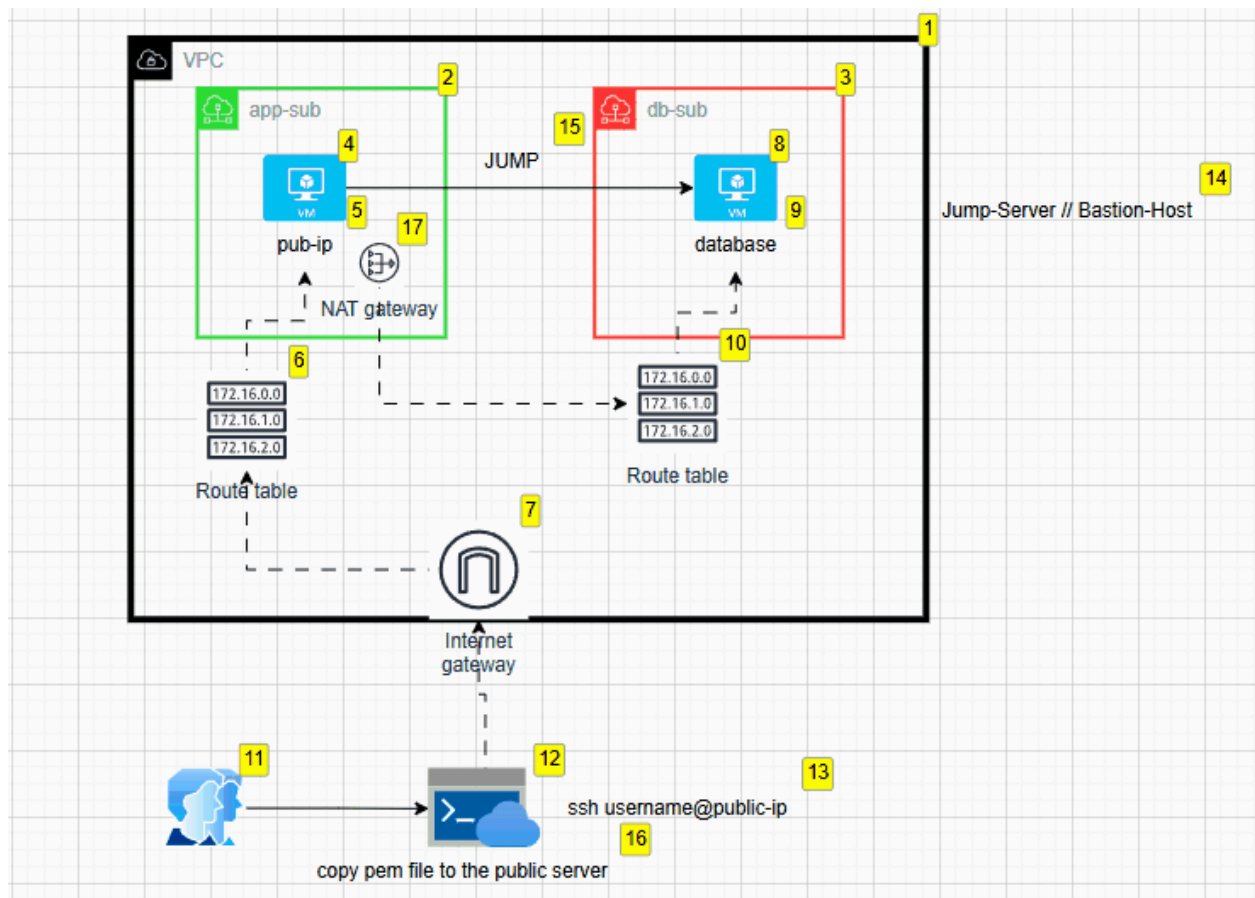

🌐 1. AWS NAT Gateway (Network Address Translation)

♦ What is it?

An **AWS NAT Gateway** allows instances in a private subnet to access the internet **outbound** (for updates, patches, etc.) **without exposing them to inbound connections** from the internet.



📌 Why NAT Gateway?

In a **public subnet**, instances have a public IP and can access the internet directly.

In a **private subnet**, instances don't have public IPs. NAT Gateway acts as a **proxy** for these private instances to **initiate** internet connections securely.

How it works:

Private Instance → NAT Gateway (in Public Subnet) → Internet

- The private instance sends traffic to the NAT Gateway
 - The NAT Gateway translates the private IP to its own **Elastic IP**
 - Response traffic comes back via the NAT Gateway
-

Use Cases:

- Private EC2 instances downloading software updates
 - Accessing S3 buckets from private subnets
 - Enabling secure outbound-only traffic
-

Pricing Note:

NAT Gateway incurs **hourly charges + data transfer costs**. For smaller workloads, you might prefer a **NAT instance** (EC2 running as a NAT proxy).

Key Points:

- Managed by AWS (highly available in 1 AZ; use multiple for redundancy)
 - Needs to be in a **public subnet**
 - Needs an **Elastic IP**
 - Route table of private subnet must send internet traffic to NAT Gateway
-

2. Bastion Host (Jump Box)

♦ What is it?

A **Bastion Host** is a **publicly accessible EC2 instance** that acts as a **secure jump point** to connect to **instances in private subnets** via SSH or RDP.

Why Bastion Host?

You shouldn't expose your internal servers to the internet. So instead of assigning public IPs to all instances, you:

- Set up **one Bastion Host** in a **public subnet**
 - SSH or RDP into the Bastion
 - From there, SSH/RDP into the **private instances**
-

How it works:

[Your Laptop]
↓ (SSH/RDP)
[Bastion Host in Public Subnet]
↓ (SSH/RDP)
[Private EC2 in Private Subnet]

Security Tips:

- Use **SSH key pairs**
 - Restrict inbound traffic to Bastion using **Security Groups** (e.g., allow only your IP)
 - Install **fail2ban** or monitoring tools
 - Use **session logging** for compliance
-

Use Cases:

- Admin access to internal/private EC2s
- Troubleshooting internal services
- Running manual tasks securely

Bastion vs NAT Gateway

Feature	NAT Gateway	Bastion Host
Purpose	Outbound internet access for private instances	Secure inbound admin access to private instances
Managed	✅ AWS-managed	❌ You manage the EC2
Cost	💰 Cost per hour + data	💰 EC2 charges
Inbound Traffic	❌ Not allowed	✅ Allowed (e.g., port 22)
Outbound Access	✅ Yes	✅ Yes
Access Protocol	TCP/IP proxy	SSH or RDP

✅ Summary

Component	Primary Use	Lives In	Key Feature
NAT Gateway	Outbound internet from private subnet	Public Subnet	Secure, no inbound access
Bastion Host	Admin access to private subnet	Public Subnet	SSH/RDP into private servers
